

RISHABH SABHARWAL

+44 7486 045605 | R.Sabharwal@sms.ed.ac.uk

<https://sabharwalrishabh.github.io/> | <https://www.linkedin.com/in/sabharwalrishabh/>

OBJECTIVE

MSc Artificial Intelligence student at the University of Edinburgh, supervised by [Prof. Jeff Pan](#), designing a benchmark to evaluate multimodal, multi-turn Deep Research Agents to assess agent reasoning, tool use, and information synthesis during long research tasks. Previously, I worked at Indian Institute of Science, under [Prof. Punit Rathore](#).

EDUCATION

MSc in Artificial Intelligence, University of Edinburgh, UK (2025–2026)

B.Tech in ECE (AI & ML), Netaji Subhas University of Technology, India (2020–2024) | **GPA: 3.68/4**

PUBLICATIONS

A.Mazumder, **Rishabh Sabharwal**, ..., P. Rathore, ‘On Convergence of Adam with Data Dependent Stepsize’ in **IEEE Transactions on AI**. ([Paper](#))

Rishabh Sabharwal, R. Samarth, P. Singh, P. Rathore, ‘DiGCT: Diffusion Guided Gaussian Context Transformer’.

SELECTED PROJECTS

LLM Post-Training on Small-Scale Models: ([Code](#))

- Fine-tuned Qwen2.5-0.5B-instruct on GSM8K using SFT and RLHF with verifiable rewards. Identified format reward over-optimization as a major issue and addressed it by using a Process Reward Model that assigns dense rewards to CoT steps, which improved the training signal quality to achieve 37% accuracy on the test set.
- Developed a **latent-space knowledge distillation** framework where the teacher (Qwen2.5-7B) generated K latent reasoning tokens transferred directly to the student. This approach avoids cognitive overload from token-level imitation and achieves measurable improvements in mathematical reasoning. Achieved an accuracy of 41% on the test set.

FastAPI Safety Gateway: ([Code](#))

- Built a guardrail around a vLLM-served Llama-3.1-8B-Instruct, combining learned and rule-based guardrails like multi-label toxicity/harm classification, prompt-injection detection, and PII detection/anonymization.
- Developed an **agentic safety loop** that rewrites risky prompts and repairs unsafe outputs through policy conditioned re-prompting.
- Enhanced deployability through user-adjustable thresholds, custom deny lists and transparent decision tracing via the Streamlit UI.

Face X-ray for Forgery Detection: ([Code](#))

- Developed a model that can locate the blending boundary (face-x-ray) in forged images. Created a set of blended images and corresponding boundaries from the FFHQ dataset.
- DeepLabV3 semantic segmentation model with EfficientNet as the encoder was finetuned to achieve 94% accuracy on the test set. A similar project to **detect deepfake images** was completed with Omdena. ([Presentation](#))

EXPERIENCE

Pre-doctoral Researcher

Indian Institute of Science

Jun 2024 - Jul 2025

Bengaluru, India

- Worked with National Payments Corporation of India to build efficient and scalable **large-scale GNNs** to detect fraudulent transactions on graphs having more than a **billion nodes**.
- Developed an efficient spectral GNN featuring a novel, parameter-free module called ReGA (Relative Gaussian Amplifier) that performs global context filtering. **Evaluated on 20 real-world graph datasets**; achieved SOTA performance, reducing parameters by an average of 55x and GFLOPs by 100x across all datasets.
- Developed DiGCT, utilizing Graph Tikhonov Regularization and graph diffusion for stronger channel attention. Increased average accuracy by **2%** on modern ConvNets with **only 20 additional parameters**.

Research Intern

Indian Institute of Science

Jan 2024 - Jun 2024

Bengaluru, India

- Proposed a network and data-dependent constant learning rate (lr) to address the issues of decaying lr schedulers and reduce the need for extensive lr search.
- Conducted empirical analysis of our proposed lr which results in faster convergence for both stochastic and deterministic variants of the Adam optimizer. **Paper accepted at IEEE Transactions on AI.** ([Link](#))

RELEVANT COURSEWORK

- **Advanced Topics in NLP:** PEFT techniques; Knowledge Distillation; Long-context Modeling; RAGs, Multi-modal LLMs; Mechanistic Interpretability and AI Safety; LLM Agents.
- **Accelerated NLP:** Classical and Neural LMs; Attention/Transformers (BERT/T5/GPT); Chain-of-Thought (CoT) Reasoning; Scaling laws; LLM Post-training.
- **Machine Learning Systems** GPU Architecture; GPU Programming (Triton); LLM Inference Optimization Techniques (KV Caching, Prefill Decode, Speculative Decoding).
- **Probabilistic Modelling and Reasoning:** Graphical Models; Exact inference (DAGs/HMMs); Causality and Decision-making under uncertainty; Variational inference and VAEs; Monte Carlo sampling.

SCHOLASTIC ACHIEVEMENTS

- Recipient of **Kotak-IISc pre-doctoral fellowship** given by Kotak-IISc AIML centre. ([website](#))
- Received the **CVSPK Fellowship for ranking in the top 5%** during my B.Tech.
- Selected as **Amazon MLSS** mentee, a 4 week program conducted by Amazon's research scientists.

ONLINE CERTIFICATIONS

- **Oxford ML Summer School:** [Representation Learning and Generative AI](#) (July 2024)
- **DeepLearning.AI:** [Deep Learning Specialization](#) (Feb 2023), [Machine Learning Specialization](#) (July 2022)
- **IBM:** [Data Science Professional Certificate](#) (Dec 2022) ([Code files](#))

SKILLS

- **Programming Languages:** Python, C++, Scala, SQL
- **Frameworks:** PyTorch, PyG, JAX, Keras, FastAPI, Streamlit
- **ML/LLM Tooling:** Hugging Face, PEFT, OpenAI-compatible APIs, Gemini API (Google Cloud/Vertex AI), RAG, LLM Tool Use
- **Agentic AI:** LangChain, LangGraph, Multi-Agent Systems
- **DevOps/Systems:** Git, GitHub, Linux, CI/CD, PyTorch DDP (NCCL)